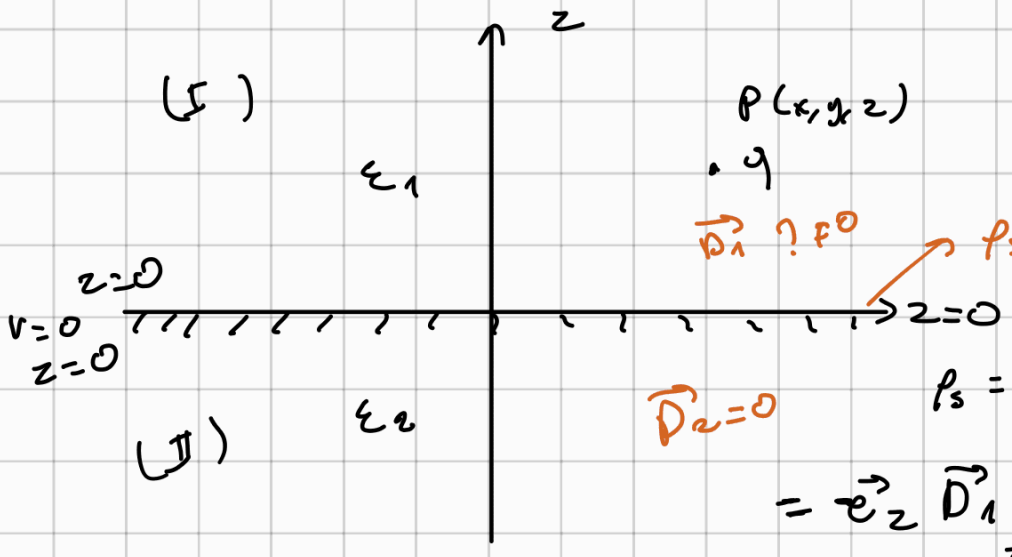
Görüntü kavramı ve yöntemi



$$\rho_s = \vec{e}_z (\vec{D}_1 - \vec{D}_2)\Big|_{z=0}$$

$$= \vec{e}_z \vec{D}_1\Big|_{z=0} = -\vec{e}_z \varepsilon_1 \text{grad } V_1\Big|_{z=0}$$

$z < 0$ bölgesi (ε_2 dielektrik sabitli bölge)

$V_2 = 0$ sabit

$\vec{E}_2 = 0$; $\vec{D}_2 = 0$

II nolu bölgede yük olmadığından $z=0$ yüzeyi üzerinde $V=0$ sabit olduğundan bölge iletkendir ve potansiyeli $V_2 = 0$ olur.

$z > 0$ bölgesi (ε_1 dielelektri sabitli bölge)

$V_1 = ?$ $\vec{E} = -\text{grad } V_1$; $\vec{D}_1 = \varepsilon_1 \vec{E}_1 = -\varepsilon_1 \text{grad } V_1$

$z > 0$ bölgesinin $\rho = q\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0) \neq 0$

I nolu bölge ise II gibi iletken bir bölge değil $V_1 \neq 0$

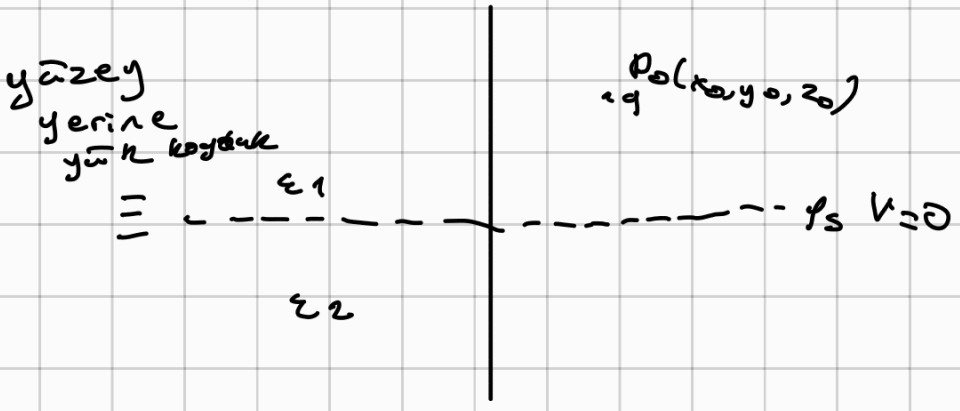
(f) V_1'in sağladığı diferansiyel denklem ve sınır koşul dikkate alınmalı.

$z > 0$ da

$$\begin{cases} \Delta V_1 = -\dfrac{\rho}{\varepsilon_1} = -q\dfrac{\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)}{\varepsilon_1} \\ V_1\big|_{z=0} = 0 \end{cases}$$

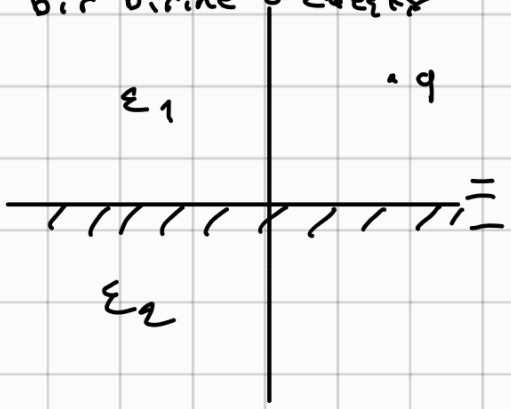
$\lim_{z\to\infty} V_1$ için $< \infty$

Bir örnek problem



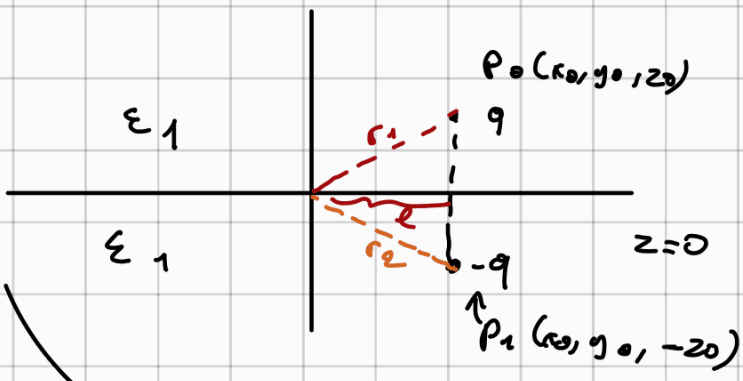
$z>0$ bölgesi bakımından

bir birine özdeştir



=

alttaki bastan biliyoruz yukarısı özdeştir



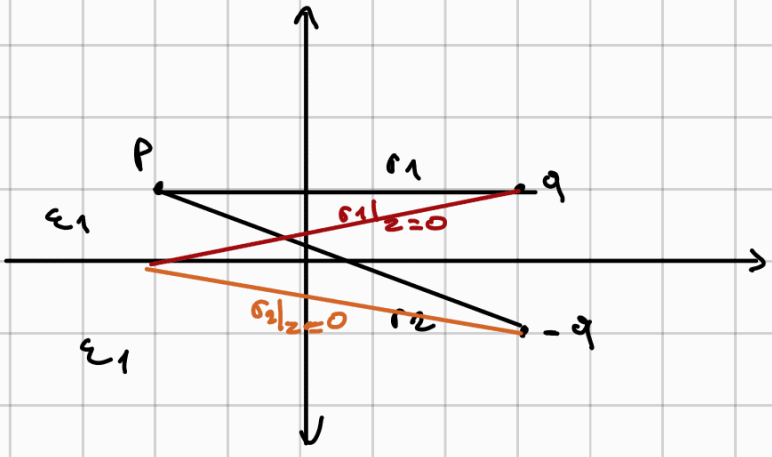
$z>0$ da $\Delta V_1 = -\dfrac{\rho_1}{\varepsilon_1} = -\dfrac{q\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)}{\varepsilon_1}$

$$V_q\big|_{z=0} = \frac{q}{4\pi\varepsilon_1 R_1} \qquad V_{-q}\big|_{z=0} = \frac{-q}{4\pi\varepsilon_1 r_2}$$

$$r_1 = r_2 = \sqrt{z_0^2 + l^2}$$

* Tüm uzay homojense Coulomb yasası geçerli

$$V\big|_{z0} = V_q\big|_{z=0} + V_{-q}\big|_{z=0} = \frac{q}{4\pi\varepsilon_1 r_1} - \frac{q}{4\pi\varepsilon_1 r_1} = V_1\big|_{z=0} = 0$$



$$V_p = \frac{q}{4\pi\varepsilon_1 r_1} - \frac{q}{4\pi\varepsilon_1 r_2}$$

$$V_p = \frac{q}{4\pi\varepsilon_1}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$V_1\big|_{z=0} = V_p\big|_{z=0} = \frac{q}{4\pi\varepsilon_1}\left(\frac{1}{r_1}\Big|_{z=0} - \frac{1}{r_2}\Big|_{z=0}\right) = 0$$

$\Rightarrow z > 0 \quad V_1 = \frac{q}{4\pi\varepsilon_1}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

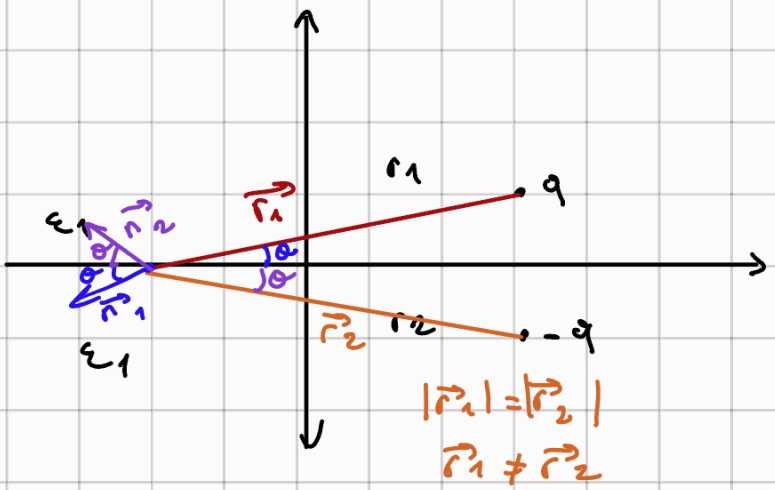
$z < 0 \quad V_2 = 0$

$\Rightarrow \vec{E}_1 = -\text{grad}\, V_1 = \frac{q}{4\pi\varepsilon_1}\frac{\vec{r}_1}{r_1^3} - \frac{q}{4\pi\varepsilon_1}\frac{\vec{r}_2}{r_2^3}$

$\vec{D}_1 = \varepsilon_1 \vec{E}_1 = \frac{q}{4\pi}\left(\frac{\vec{r}_1}{r_1^3} - \frac{\vec{r}_2}{r_2^3}\right)$

$$\rho_s\big|_{z=0} = \vec{e}_z\,(\vec{D}_1 - \vec{D}_2)\big|_{z=0} = \vec{e}_z \cdot \vec{D}_1\big|_{z=0}$$

$$\rho_s = \frac{q}{4\pi}\,\vec{e}_z\left(\frac{\vec{r}_1}{r_1^3}\Big|_{z=0} - \frac{\vec{r}_2}{r_2^3}\Big|_{z=0}\right)$$



$|\vec{r}_1| = |\vec{r}_2|$

$\vec{r}_1 \neq \vec{r}_2$

$\vec{r}_1\big|_{z=0} = r_1 \cdot \vec{n}_1$

$\vec{r}_1 = r_1\big|_{z=0}\; n_1\big|_{z=0} = r_1[\cos\theta\,\vec{e}_z - \sin\theta\,\vec{e}_z]$

$\vec{r}_2\big|_{z=0} = r_2\big|_{z=0}\; n_2\big|_{z=0} = r_2[\cos\theta\,\vec{e}_z + \sin\theta\,\vec{e}_z]$

$r_1 = r_2 = r$

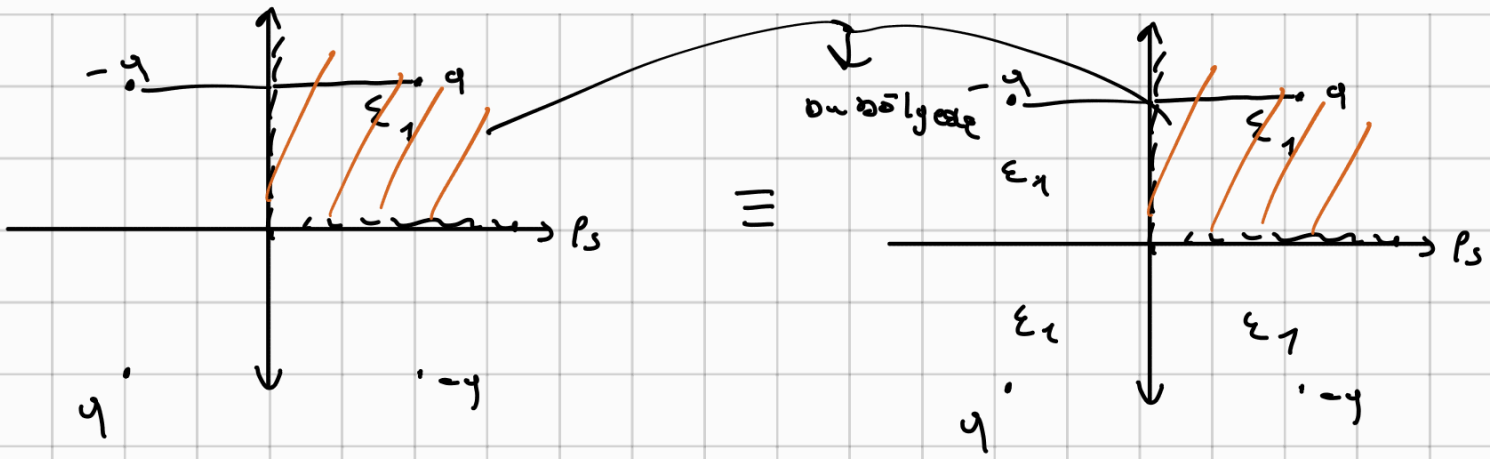
$$\Rightarrow \rho_s = \frac{q}{4\pi}\,\vec{e}_z\left(\frac{\vec{r}_1|_{z=0}}{\underbrace{(r_1|_{z=0})^3}_{r^3}} - \frac{\vec{r}_2|_{z=0}}{\underbrace{(r_2|_{z=0})^3}_{r^3}}\right) = \frac{q}{4\pi}\frac{\vec{e}_z}{r^3}\left(\vec{r}_1|_{z=0} - \vec{r}_2|_{z=0}\right)$$

$$= \frac{q}{4\pi r^3}\,\vec{e}_z\left[r\cos\theta\,\vec{e}_z - r\sin\theta\,\vec{e}_z - r\cos\theta\,\vec{e}_z - r\sin\theta\,\vec{e}_z\right]$$

$$= \frac{-q\sin\theta}{2\pi r^2} \qquad \text{ve} \quad \sin\theta = \frac{\text{Karşı}}{\text{hipo}} = \frac{z_0}{r}$$

yüzeyin üzerindeki yük 0 değildir ve değişkendir

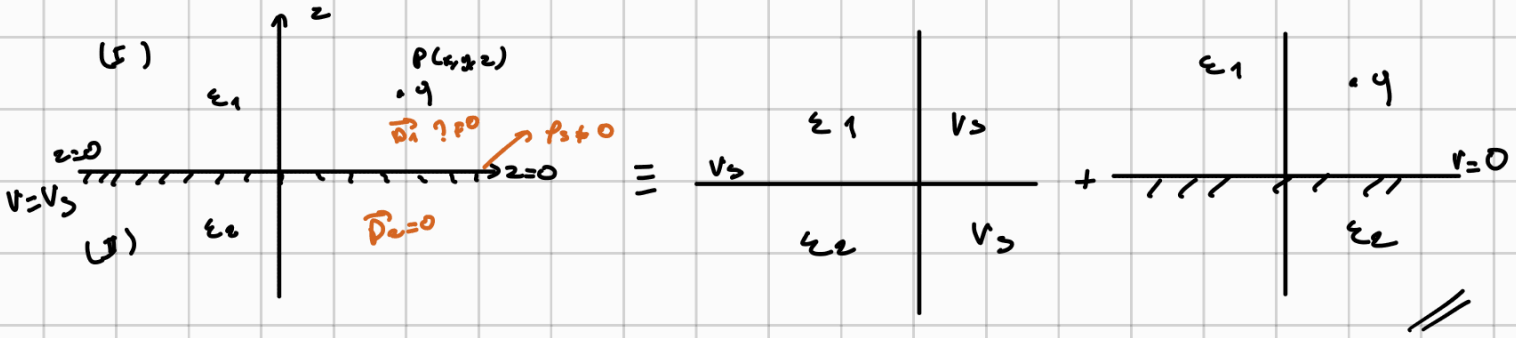
$$= \frac{-q\,z_0}{2\pi r^3} \quad (C/m^2)$$

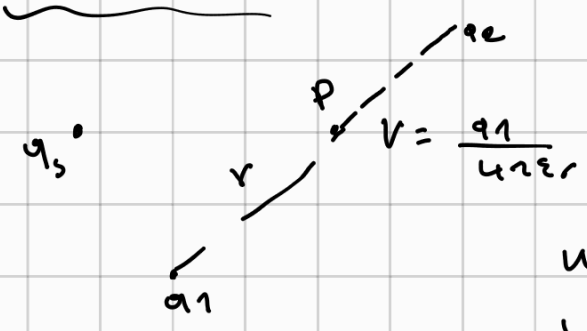




$n_i = \frac{2\pi}{\alpha} - 1$ görüntü yük sayısı

tam sayı gelmezse sonsuz görüntü var demektir.





q_1 başta vardı

q_2 uzaktan getirildi:

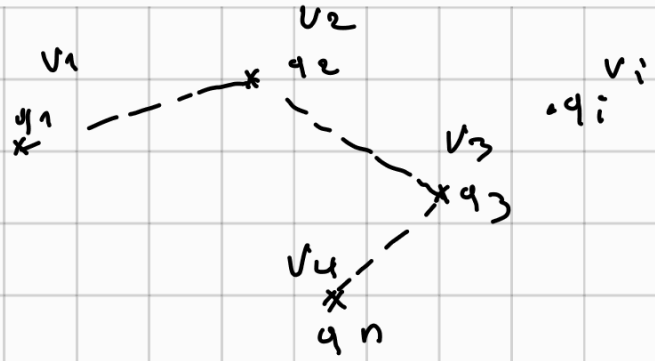
$$W = q_2(V_1 - \frac{V_\infty}{0}) = q_2 \frac{q_1}{4\pi\varepsilon r}$$

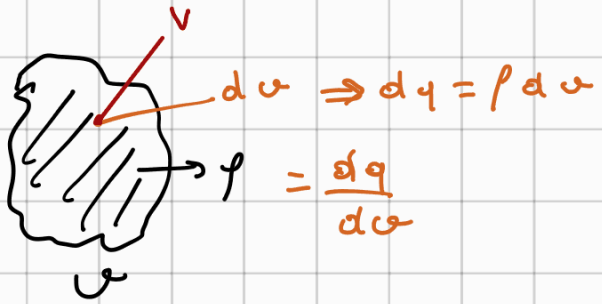
$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V_i$$

$$= \frac{1}{2}[q_1 V_1 + q_2 V_2 + \cdots + q_n V_n]$$

$$W_1 = W_{13} + W_{23} + W_{12}$$

$$= \frac{q_3 q_1}{4\pi\varepsilon r_{13}} + \frac{q_3 \cdot q_2}{4\pi\varepsilon r_{23}} + \frac{q_1 \cdot q_2}{4\pi\varepsilon r_{12}}$$





$$dV = \frac{1}{2}\rho\, dv \cdot V$$

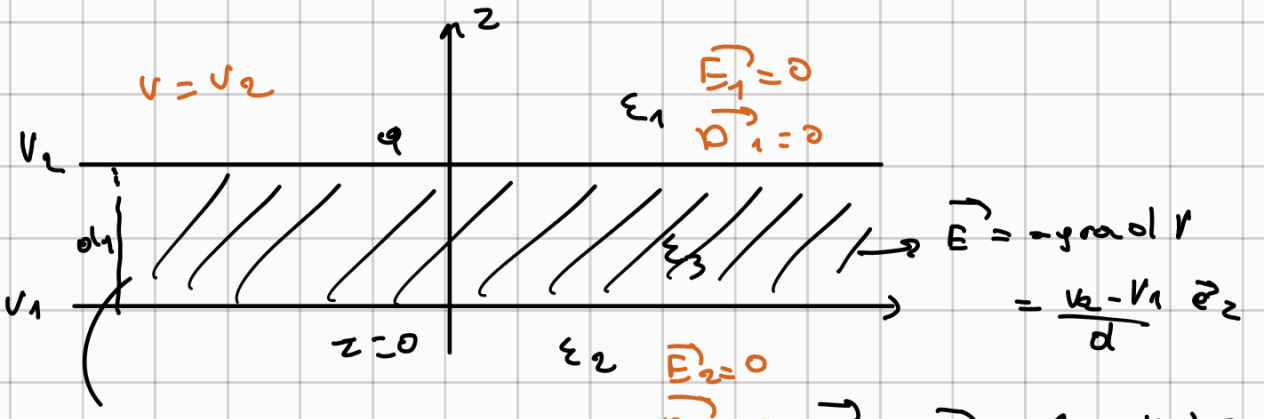
$$\Rightarrow W = \frac{1}{2}\int_v \rho V\, dv$$

$$= \frac{1}{2}\int \rho V\, dv \qquad = \frac{1}{2}\int \underbrace{\rho}_{\operatorname{div}\vec{D}} V\, dv$$

$$\vec{E} = -\operatorname{grad} V = -\frac{\partial V}{\partial n}\vec{n}$$

$$= \frac{1}{2}\int \rho V\, dv = \frac{1}{2}\int \vec{E}\vec{D}\, dv = \frac{1}{2}\int \varepsilon |\vec{E}|^2 dv$$

($\vec{D} = \varepsilon\vec{E}$)



$\Delta V = 0$

$V = Az + B$

$B = V|_{z=0} = V_1 \qquad V|_{z=d} = V_2 = Ad + V_1$

$$A = \frac{V_2 - V_1}{d}$$

$$W = \frac{1}{2}\int \varepsilon |\vec{E}|^2 \, d\tau$$

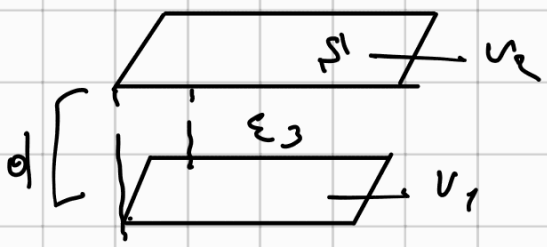
$$W = = \frac{1}{2}\int_{\vartheta_1} \varepsilon_1 |\vec{E}_1|^2 d\tau_1 + \frac{1}{2}\int_{\vartheta_3} \varepsilon_3 |\vec{E}_3|^3 d\tau + \frac{1}{2}\int_{V_2} \varepsilon_2 |\vec{E}_2|^2 d\tau_2$$

$\vec{E}_2 \to 0$, 0

$$= \frac{\varepsilon_3}{2} \frac{(V_2 - V_1)^2}{d^2} \int_{\vartheta_3} d\tau_3 = \frac{\varepsilon_3}{2} \frac{(V_2 - V_1)^2}{d} S' d$$

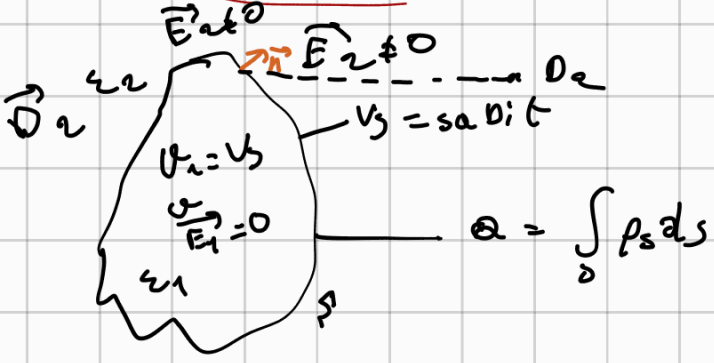
$S'd$

$$= \frac{\varepsilon_3}{2d} S' (V_2 - V_1)^2$$



← kapasite

Kapasite problemi



$$\vec{D}_2 = \varepsilon_2 (-\text{grad } V_2) = \varepsilon_2 \, \text{grad } V_2$$

$$\rho_s = \vec{n} (\vec{D}_2 - \vec{D}_1)\Big|_S = \vec{n} \cdot \vec{D}_2 \Big|_S$$

$$= -\vec{n} \, \varepsilon_2 \, \text{grad } V_2 \Big|_S = -\varepsilon_2 \frac{\partial V_2}{\partial n}\Big|_S$$

$$\frac{\partial V_2}{\partial n} = -\text{grad } V_2 \cdot \vec{n}\Big|_S$$

$$= -\varepsilon_2 \int_S \frac{\partial V_2}{\partial n} \, ds$$

$V_2 = V_s \; \Phi(x,y,z)$

$V_2|_s = V_s \Phi(x,y,z)|_s = V_s$

$\Rightarrow V_s \Phi(x,y,z)|_s = V_s$

$\Rightarrow \Phi(x,y,z)|_s = 1$ geometriye bağlı fonksiyondur.

$\Delta V_2 = V_s \Delta \Phi = 0 = -\frac{\rho}{\varepsilon_2} = 0$

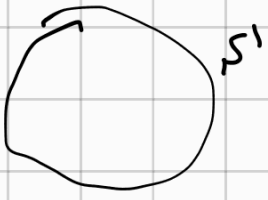
$\Delta \Phi = 0$

$\Phi|_s = 1$

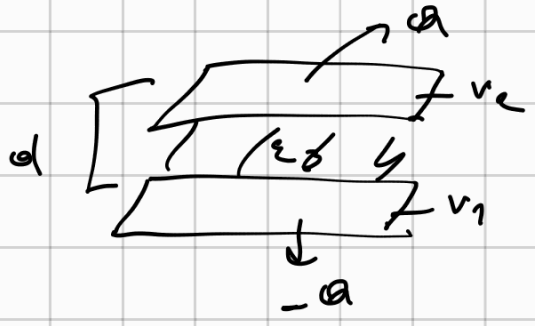
$Q = -\varepsilon_2 V_s \int_S \frac{\partial \Phi}{\partial n} dS$

$\frac{Q}{V_s} = C = -\varepsilon_2 \int_{S'} \frac{\partial \Phi}{\partial \hat{n}} dS$

iletkenin yapısı



$\Rightarrow Q = CV_s$



$Q = C.(V_2 - V_1)$

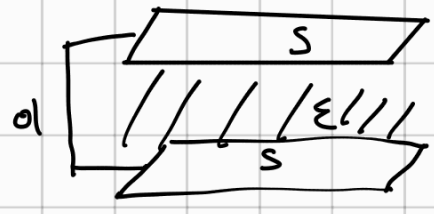
$-Q = C.(V_1 - V_2)$

$$W = \frac{1}{2} Q (V_2 - V_1) = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} C (V_2 - V_1)^2$$

$$= \frac{\varepsilon_2}{2d} S^1 (V_2 - V_1)^2$$

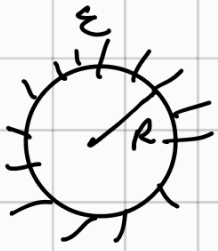
$$\Rightarrow C = \frac{\varepsilon_3 S}{d}$$

$\Rightarrow$



$$C = \frac{\varepsilon \cdot S}{d} \qquad \frac{(\text{Farad}/m)\ (m^2)}{m} = \text{Farad}$$

Bir kürenin kapasitesi



$$C = 4\pi \varepsilon R$$

$$V = \frac{V_s \cdot R}{r} = V_s \Phi \Rightarrow \Phi = \frac{R}{r}$$

$$ds = r^2 \sin\theta \, d\theta \, d\phi$$

$$C = -\varepsilon \int_{S_R} \frac{-R}{r^2} = C \varepsilon R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \sin\theta \, d\theta \, d\phi$$

$$= 4\pi \varepsilon R$$

silindir için

$$V = A \ln \rho$$

$$V = A \ln R = V_s \Rightarrow A = \frac{V_s}{\ln r}$$

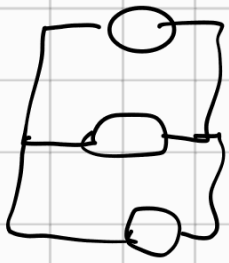
$$\Rightarrow V = \frac{V_s}{\ln R} \ln \rho = V_s \Phi \Rightarrow \Phi = \frac{\ln \rho}{\ln R}$$

$dn = dq$

$$= -\varepsilon \int_{S^1} \frac{\partial \Phi}{\partial n} ds = -\varepsilon \int \frac{\ln \rho}{\ln R} \underbrace{\rho \, d\phi \, dz}_{ds}$$

$$= -\varepsilon \int_{\phi=0}^{2\pi} \int_{0}^{\ell} \frac{1}{\ln R} \frac{1}{\rho} \rho \, d\phi \, dz = \frac{-\varepsilon}{\ln R} 2\pi \ell$$

$$= \frac{2\pi\varepsilon\ell}{-\ln R} = \frac{2\pi\varepsilon\ell}{\ln(1/R)}$$



$$C = C_1 + C_2 + C_3$$